

1. Given, $f(x) = \int_0^6 (e^x + 2x) dx$ Trapezoidal

Here, upper limit is, $b=6$ and lower limit, $a=0$ & no of sub-intervals, $n=6$

Now, $h = \frac{6-0}{6} = 1$

The values of the function y at each sub-interval are given below:

x	$x_0 = 0$	$x_1 = 1$	$x_2 = 2$	$x_3 = 3$	$x_4 = 4$	$x_5 = 5$	$x_6 = 6$
y	$y_0 = 1$	$y_1 = 4.7183$	$y_2 = 11.3891$	$y_3 = 25.0855$	$y_4 = 62.5982$	$y_5 = 158.9132$	$y_6 = 416.4288$

We know,

$$\int_{x_0}^{x_0+nh} y dx = \frac{h}{2} \left[(y_0 + y_n) + 2(y_1 + y_2 + y_3 + y_4 + y_5) \right]$$

$$= \frac{1}{2} \left[416.4288 + 2(262.2043) \right]$$

$$= \frac{1}{2} \left[416.4288 + 524.4086 \right]$$

$$= 470.4187$$

Simpson's 1/3

We know,

$$\int_{x_0}^{x_0+nh} y dx = \frac{h}{3} \left[(y_0 + y_n) + 4(y_1 + y_3 + y_5 + \dots + y_{n-1}) + 2(y_2 + y_4 + y_6 + \dots + y_{n-2}) \right]$$

For $n=6$ we get,

$$\int_0^6 (e^x + 2x) dx = \frac{1}{3} \left[(1 + 415.4288) + 4(4.7183 + 25.0855 + 158.4132) + 2(11.3891 + 62.5982) \right]$$

$$= 439.0905$$

Simpson's 3/8

We know,

$$\int_{x_0}^{x_0+nh} y dx = \frac{3h}{8} \left[(y_0 + y_n) + 3(y_1 + y_2 + y_4 + y_5 + y_7 + \dots + y_{n-2} + y_{n-1}) + 2(y_3 + y_6 + y_9 + \dots + y_{n-3}) \right]$$

For $n=6$ we get,

$$\int_0^6 (e^x + 2x) dx = \frac{3 \times 1}{8} \left[(1 + 415.4288) + 3(4.7183 + 11.3891 + 62.5982 + 158.4132) + 2(25.0855) \right]$$

$$= 441.7336$$

Weedle's

We know,

$$\int_a^b f(x) dx = \frac{3h}{10} \left[(y_0 + y_2 + y_4 + y_6 + y_8 + \dots + y_n) + 5(y_1 + y_3 + y_5 + \dots + y_{n-1}) + (y_3 + y_6 + y_9 + \dots + y_{n-3}) \right]$$

For $n=6$ we get,

$$\int_0^6 (e^{x+2x}) dx = \frac{3 \times 1}{10} \left[(1 + 11.3891 + 62.5982 + 415.9288) + 5(9.7183 + 25.0855 + 158.4132) + 25.0855 \right]$$

$$= 436.9760$$

Exact value: $\int_0^6 (e^x + 2x) dx$

$$= \left[e^x + 2 \cdot \frac{x^2}{2} \right]_0^6$$

$$= (e^6 - e^0) + 36$$

$$= 438.4288$$

percentage error for trapezoidal = $\left| \frac{438.4288 - 420.4187}{438.4288} \right| \times 100$

$$= 7.2965\%$$

percentage error for Simpson's $1/3$ = $\left| \frac{438.4288 - 439.0905}{438.4288} \right| \times 100$

$$= 0.1509\%$$

percentage error for Simpson's $3/8$ = $\left| \frac{438.4288 - 441.7336}{438.4288} \right| \times 100$

$$= 0.7538\%$$

percentage error for wedge = $\left| \frac{438.4288 - 436.9760}{438.4288} \right| \times 100$

$$= 0.3314\%$$

From the percentage error it's clear that Simpson's $1/3$ gives the most accurate result that is 0.1509% & Simpson's $3/8$ gives the lowest accurate result that is 0.7538% .

2. Given,

The exponential equation, $y = ae^{bx}$ — (i)

Taking $\ln$ on each side of (i) we get,

$$\ln y = \ln(ae^{bx})$$

$$\Rightarrow \ln y = \ln a + \ln e^{bx}$$

$$\Rightarrow \ln y = \ln a + bx \ln e \longrightarrow (ii)$$

Let,

$$Y = \ln y, A = \ln a, B = b \ln e$$

So, the equation (ii) becomes,

$$Y = A + Bx$$

The normal equations are,

$$\sum Y = nA + B \sum x \longrightarrow (iii)$$

$$\sum xY = ~~A \sum x~~ A \sum x + B \sum x^2 \longrightarrow (iv)$$

Here, the number of data points of x , $n=4$
 Calculation for finding the coefficient of a & b we form the following table

x	y	$Y = \ln y$	xY	x^2
1	3	1.0986	1.0986	1
2	5	1.6094	3.2188	4
3	7	1.9459	5.8372	9
4	10	2.3026	9.2104	16
$\Sigma x = 10$		$\Sigma Y = 6.9565$	$\Sigma xY = 19.365$	$\Sigma x^2 = 30$

putting the values in eqn (iii) & (iv) we get,

$$4A + 10B = 6.9565 \quad \text{--- (v)}$$

$$10A + 30B = 19.3655 \quad \text{--- (vi)}$$

$9 \times (vi) - 10 \times (v)$ we get,

$$90A + 120B = 77.462$$

$$90A + 100B = 69.565$$

$$\begin{array}{r} \ominus \\ \ominus \end{array} \quad \begin{array}{r} \ominus \\ \ominus \end{array}$$

$$20B = 7.897$$

$$\therefore B = 0.3949$$

putting the value in eqn (v) we get,

$$4A + 3.949 = 6.9565$$

$$\therefore A = 0.7519$$

Now,

$$A = \ln a$$

$$\Rightarrow a = e^A$$

$$\Rightarrow a = e^{0.7519}$$

$$\therefore a = 2.1210$$

$$B = \ln b$$

$$\Rightarrow b = 0.3949$$

$$\therefore b = 0.3949$$

Putting the values in eqn (i) we get,

$$y = (2.1210) e^{(0.3949)x}$$

For $x=6$,

$$y = (2.1210) e^{0.3949 \times 6}$$

$$\therefore y = 22.6756$$

3. Let,

The equation of polynomial, $y = a + bx + cx^2$ — (i)

The normal equations are,

$$\sum y = na + b \sum x + c \sum x^2 \text{ — (ii)}$$

$$\sum xy = a \sum x + b \sum x^2 + c \sum x^3 \text{ — (iii)}$$

$$\sum x^2 y = a \sum x^2 + b \sum x^3 + c \sum x^4 \text{ — (iv)}$$

Here, the number of data points of x , $n = 4$
 calculation for finding the coefficient of a & b we form the following table,

x	y	xy	x^2y	x^2	x^3	x^4
1	2	2	2	1	1	1
2	4	8	16	4	8	16
3	6	18	54	9	27	81
4	9	36	144	16	64	256
$\sum x = 10$	$\sum y = 21$	$\sum xy = 64$	$\sum x^2y = 216$	$\sum x^2 = 30$	$\sum x^3 = 100$	$\sum x^4 = 354$

Putting these values in eqn (ii), (iii) & (iv) we get,

$$4a + 10b + 30c = 21 \text{ — (v)}$$

$$10a + 30b + 100c = 64 \text{ — (vi)}$$

$$30a + 100b + 354c = 216 \text{ — (vii)}$$

Now,

$4 \times (vi) - 10 \times (v)$ we get,

$$40a + 120b + 400c = 256$$

$$40a + 100b + 300c = 210$$

$$\begin{array}{r} 40a + 120b + 400c = 256 \\ - (40a + 100b + 300c = 210) \\ \hline 20b + 100c = 46 \end{array}$$

$$\Rightarrow 10b + 50c = 23$$

$$\Rightarrow b = \frac{23 - 50c}{10} \text{ — (viii)}$$

$4 \times (vii) - 30 \times (v)$ we get,

$$120a + 400b + 1416c = 864$$

$$120a + 300b + 900c = 630$$

$$\begin{array}{r} 120a + 400b + 1416c = 864 \\ - (120a + 300b + 900c = 630) \\ \hline 100b + 516c = 234 \end{array}$$

$$\Rightarrow 50b + 258c = 117$$

$$\Rightarrow b = \frac{117 - 258c}{50} \text{ — (ix)}$$

$$\Rightarrow \frac{23-50c}{10} = \frac{117-258c}{50}$$

$$\Rightarrow 1170 - 2580c = 1150 - 2500c$$

$$\Rightarrow -80c = -20$$

$$\Rightarrow c = \frac{1}{4}$$

$$\therefore c = 0.25$$

putting the value in eqn (viii) we get,

$$b = \frac{23 - (50 \times 0.25)}{10} = 1.05$$

putting these values in eqn (v) we get,

$$9a + 10.5 + 7.5 = 21$$

$$\therefore a = 0.75$$

putting these values in eqn (i) we get,

$$y = 0.75 + 1.05x + 0.25x^2$$

For $x=5$ the eqn becomes,

$$\begin{aligned} y &= 0.75 + (1.05 \times 5) + 0.25 \times 25 \\ &= 12.25 \end{aligned}$$

9. Given,

$$10 \frac{dy}{dx} = x^2 + y^2 \Rightarrow f(x, y) = \frac{dy}{dx} = \frac{x^2 + y^2}{10}$$

Here, $x_0 = 0$, $y_0 = 1$

Taking $h = 0.5$, $x_1 = 0.5$

By Runge-Kutta order-4 method,

$$y_{n+1} = y_n + \frac{1}{6} [K_1 + 2K_2 + 2K_3 + K_4] \quad \text{--- (i)}$$

where,

$$K_1 = hf(x_n, y_n), \quad K_2 = hf\left(x_n + \frac{h}{2}, y_n + \frac{K_1}{2}\right)$$

$$K_3 = hf\left(x_n + \frac{h}{2}, y_n + \frac{K_2}{2}\right), \quad K_4 = hf(x_n + h, y_n + K_3)$$

Step-1: putting $n=0$

$$K_1 = hf(x_0, y_0)$$

$$= 0.5 \left[\frac{0^2 + 1^2}{10} \right] = 0.05$$

$$K_2 = hf\left(x_0 + \frac{h}{2}, y_0 + \frac{K_1}{2}\right)$$

$$= 0.5 \left[\frac{0 + \frac{0.5}{2}, 1 + \frac{0.05}{2}}{10} \right]$$

$$= 0.5 \left[\frac{0.25, 1.025}{10} \right]$$

$$= 0.5 \left[\frac{(0.25)^2 + (1.025)^2}{10} \right]$$

$$= 0.0557$$

$$K_3 = hf\left(x_0 + \frac{h}{2}, y_0 + \frac{K_2}{2}\right)$$

$$= 0.5 \left[\frac{0 + \frac{0.5}{2}, 1 + \frac{0.0557}{2}}{10} \right]$$

$$= 0.5 \left[\frac{0.25, 1.0279}{10} \right]$$

$$= 0.5 \left[\frac{(0.25)^2 + (1.0279)^2}{10} \right]$$

$$= 0.05595$$

$$K_4 = hf(x_0 + h, y_0 + K_3)$$

$$= 0.5 \left[\frac{0 + 0.5, 1 + 0.05595}{10} \right]$$

$$= 0.5 \left[\frac{0.5, 1.05595}{10} \right]$$

$$= 0.5 \left[\frac{(0.5)^2 + (1.05595)^2}{10} \right]$$

$$= 0.0683$$

$$\begin{aligned}\therefore y_1 &= y(0.5) = y_0 + \frac{1}{6} [k_1 + 2k_2 + 2k_3 + k_4] \\ &= 1 + \frac{1}{6} [0.05 + 2 \times 0.0557 + 2 \times 0.05595 + 0.0683] \\ &= 1.0562\end{aligned}$$

Step-2: Putting $n=1$

$$\begin{aligned}k_1 &= hf(x_1, y_1) \\ &= 0.5 f[0.5, 1.0562] \\ &= 0.5 \left[\frac{(0.5)^2 + (1.0562)^2}{10} \right] \\ &= 0.0683\end{aligned}$$

$$\begin{aligned}k_2 &= hf\left(x_1 + \frac{h}{2}, y_1 + \frac{k_1}{2}\right) \\ &= 0.5 f\left[0.5 + \frac{0.5}{2}, 1.0562 + \frac{0.0683}{2}\right] \\ &= 0.5 f[0.75, 1.0904] \\ &= 0.5 \left[\frac{(0.75)^2 + (1.0904)^2}{10} \right] \\ &= 0.0876\end{aligned}$$

$$\begin{aligned}k_3 &= hf\left(x_1 + \frac{h}{2}, y_1 + \frac{k_2}{2}\right) \\ &= 0.5 f\left[0.5 + \frac{0.5}{2}, 1.0562 + \frac{0.0876}{2}\right] \\ &= 0.5 f[0.75, 1.1] \\ &= 0.5 \left[\frac{(0.75)^2 + (1.1)^2}{10} \right] \\ &= 0.0886\end{aligned}$$

$$\begin{aligned}k_4 &= hf(x_1 + h, y_1 + k_3) \\ &= 0.5 f[0.5 + 0.5, 1.0562 + 0.0886] \\ &= 0.5 f[1, 1.1448] \\ &= 0.5 \left[\frac{1^2 + (1.1448)^2}{10} \right] \\ &= 0.1155\end{aligned}$$

$$y(1.0) =$$

$$\begin{aligned}\therefore y_2 &= y_1 + \frac{1}{6} [k_1 + 2k_2 + 2k_3 + k_4] \\ &= 1.0562 + \frac{1}{6} [0.0683 + 2 \times 0.0876 + 2 \times 0.0886 + 0.1155] \\ &= 1.1456\end{aligned}$$

step-3: Putting $n=2$

$$K_1 = hf(x_2, y_2)$$

$$= 0.5 f[1.00, 1.1456]$$

$$= 0.5 \left[\frac{(1.00)^2 + (1.1456)^2}{10} \right]$$

$$= 0.1156$$

$$K_2 = hf(x_2 + \frac{h}{2}, y_2 + \frac{K_1}{2})$$

$$= 0.5 f[1.00 + \frac{0.5}{2}, 1.1456 + \frac{0.1156}{2}]$$

$$= 0.5 f[1.25, 1.2034]$$

$$= 0.5 \left[\frac{(1.25)^2 + (1.2034)^2}{10} \right]$$

$$= 0.1505$$

$$\therefore y_3 = y(1.5) = y_2 + \frac{1}{6} [K_1 + 2K_2 + 2K_3 + K_4] = 1.1456 + \frac{1}{6} [0.1156 + 2 \times 0.1505 + 2 \times 0.1527 + 0.1968]$$

$$K_3 = hf(x_2 + \frac{h}{2}, y_2 + \frac{K_2}{2})$$

$$= 0.5 f[1.25, 1.1456 + \frac{0.1156}{2}]$$

$$= 0.5 f[1.25, 1.2209]$$

$$= 0.5 \left[\frac{(1.25)^2 + (1.2209)^2}{10} \right]$$

$$= 0.1527$$

$$K_4 = hf(x_2 + h, y_2 + K_3)$$

$$= 0.5 f[1.00 + 0.5, 1.1456 + 0.1527]$$

$$= 0.5 f[1.5, 1.2983]$$

$$= 0.5 \left[\frac{(1.5)^2 + (1.2983)^2}{10} \right]$$

$$= 0.1968$$

$$= 1.2482$$

Step-4: Putting $n=3$

$$\begin{aligned} K_1 &= hf(x_3, y_3) \\ &= 0.5 \left[\frac{(1.5)^2 + (1.2982)^2}{10} \right] \\ &= 0.1968 \end{aligned}$$

$$\begin{aligned} K_2 &= hf(x_3 + \frac{h}{2}, y_3 + \frac{K_1}{2}) \\ &= 0.5 f\left(1.5 + \frac{0.5}{2}, 1.2982 + \frac{0.1968}{2}\right) \\ &= 0.5 f(1.75, 1.3971) \\ &= 0.5 \left[\frac{(1.75)^2 + (1.3971)^2}{10} \right] \\ &= 0.2507 \end{aligned}$$

$$\begin{aligned} K_3 &= hf(x_3 + \frac{h}{2}, y_3 + \frac{K_2}{2}) \\ &= 0.5 f(1.75, 1.4241) \\ &= 0.5 \left[\frac{(1.75)^2 + (1.4241)^2}{10} \right] \\ &= 0.2545 \end{aligned}$$

$$\begin{aligned} K_4 &= hf(x_3 + h, y_3 + K_3) \\ &= 0.5 f(2, 1.5532) \\ &= 0.5 \left[\frac{4 + (1.5532)^2}{10} \right] \\ &= 0.3206 \end{aligned}$$

$$\begin{aligned} y_4 &= y(2.0) = y_3 + \frac{1}{6} [K_1 + 2K_2 + 2K_3 + K_4] = 1.2982 + \frac{1}{6} [0.1968 + 2 \times 0.2507 + 2 \times 0.2545 + 0.3206] \\ &= 1.5533 \end{aligned}$$

~~$y_1 = y(0)$~~

$\therefore y(0) = 1$

$y(0.5) = 1.0562$

$y(1.0) = 1.1456$

$y(1.5) = 1.2982$

$y(2.0) = 1.5533$

5. Given,

$$\left. \begin{aligned} x+2y+5z &= 20 \\ 5x+2y+z &= 12 \\ x+4y+2z &= 15 \end{aligned} \right\} \text{--- (i)}$$

Here, $A = \begin{bmatrix} 1 & 2 & 5 \\ 5 & 2 & 1 \\ 1 & 4 & 2 \end{bmatrix}$, $X = \begin{pmatrix} x \\ y \\ z \end{pmatrix}$, $B = \begin{pmatrix} 20 \\ 12 \\ 15 \end{pmatrix}$

Here, the coefficient matrix A is not strictly diagonally dominant because,

$$|a_{11}| < |a_{12}| + |a_{13}| = 4 < 7$$

$$|a_{22}| < |a_{21}| + |a_{23}| = 2 < 6$$

$$|a_{33}| < |a_{31}| + |a_{32}| = 2 < 5$$

Re-arranging the system (i) as follows to make strictly diagonally dominant,

$$A = \begin{bmatrix} 5 & 2 & 1 \\ 1 & 4 & 2 \\ 1 & 2 & 5 \end{bmatrix} \text{ and } B = \begin{pmatrix} 12 \\ 15 \\ 20 \end{pmatrix}$$

The given system (i) becomes,

$$\left. \begin{aligned} 5x+2y+z &= 12 \\ x+4y+2z &= 15 \\ x+2y+5z &= 20 \end{aligned} \right\} \text{--- (ii)}$$

From eqn (ii) we get,

$$x = \frac{1}{5} (12 - 2y - z)$$

$$y = \frac{1}{4} (15 - x - 2z)$$

$$z = \frac{1}{5} (20 - x - 2y)$$

Taking initial values: $x=0$, $y=0$ & $z=0$

Iteration-1: putting $k=1$

$$x^{(1)} = \frac{1}{5} (12 - 0 - 0) = 2.4$$

$$y^{(1)} = \frac{1}{4} (15 - 2.4 - 0) = 3.15$$

$$z^{(1)} = \frac{1}{5} (20 - 2.4 - 2 \times 3.15) = 2.26$$

Iteration-2: Putting $k=2$

$$x^{(2)} = \frac{1}{5} (12 - 2 \times 3.15 - 2.26) = 0.688$$

$$y^{(2)} = \frac{1}{4} (15 - 0.688 - 2 \times 2.26) = 2.448$$

$$z^{(2)} = \frac{1}{5} (20 - 0.688 - 2 \times 2.448) = 2.8832$$

Iteration-3: Putting $k=3$

$$x^{(3)} = \frac{1}{5} (12 - 2 \times 2.448 - 2.8832) = 0.8442$$

$$y^{(3)} = \frac{1}{4} (15 - 0.8442 - 2 \times 2.8832) = 2.0974$$

$$z^{(3)} = \frac{1}{5} (20 - 0.8442 - 2 \times 2.0974) = 2.9922$$

Iteration-4: putting $k=4$

$$x^{(4)} = \frac{1}{5} (12 - 2 \times 2.0974 - 2.9922) = 0.9626$$

$$y^{(4)} = \frac{1}{4} (15 - 0.9626 - 2 \times 2.9922) = 2.0133$$

$$z^{(4)} = \frac{1}{5} (20 - 0.9626 - 2 \times 2.0133) = 3.0022$$

Iteration-5: putting $k=5$

$$x^{(5)} = \frac{1}{5} (12 - 2 \times 2.0133 - 3.0022) = 0.9942$$

$$y^{(5)} = \frac{1}{4} (15 - 0.9942 - 2 \times 3.0022) = 2.0004$$

$$z^{(5)} = \frac{1}{5} (20 - 0.9942 - 2 \times 2.0004) = 3.001$$

Iteration - 6: Putting $k=6$

$$x^{(6)} = \frac{1}{5} (12 - 2 \times 2.0004 - 3.001) = 0.9996$$

$$y^{(6)} = \frac{1}{4} (15 - 0.9996 - 2 \times 3.001) = 1.9996$$

$$z^{(6)} = \frac{1}{5} (20 - 0.9996 - 2 \times 1.9996) = 3.0002$$

Iteration - 7: Putting $k=7$

$$x^{(7)} = \frac{1}{5} (12 - 2 \times 1.9996 - 3.0002) = 1.0001$$

$$y^{(7)} = \frac{1}{4} (15 - 1.0001 - 2 \times 3.0002) = 1.9999$$

$$z^{(7)} = \frac{1}{5} (20 - 1.0001 - 2 \times 1.9999) = 3.0000$$

After seventh iteration the difference between iteration-6 and iteration-7 is very negligible.

∴ The solution of the given system is: $x=1.0001, y=1.9999, z=3.0000$